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Pediatric Limb Amputation: Aspects of Coping and Psychotherapeutic Intervention

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ABSTRACT: This paper addresses the assessment of and intervention with pediatric patients undergoing limb amputation. A multicomponent treatment package is advocated including the use of play and cognitive-behavioral strategies to enhance coping and adjustment to the loss of a limb.

KEY WORDS: pediatric amputation; limb loss; coping; play therapy.

Limb amputation in the pediatric patient, though relatively infrequent, can present unique stressors for both the child and family members.^{1,3} While the majority of literature on the topic is confined to orthopedic aspects, frequent reference is made to the psychic trauma entailed, such as that by Letts⁴: "To the unprepared child, the loss of a considerable segment of body tissue is often psychologically more traumatic than is the actual surgery" (pp. 605).

The etiology of limb loss in the child or adolescent may be due to congenital factors or acquired secondary to the effects of disease or trauma. There are indications that the etiology of the limb loss may have differential effects on the child and family's functioning.⁵ Congenital limb loss may be associated with a much longer period of preparation for the limitations associated with amputation, while traumatic loss of limb places immediate demands on the adaptive skills and resources of the child and family. When amputation is ac-

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quired secondary to disease, factors such as ongoing treatment for the underlying disease, the possibility of recurrence or of amputation decreasing the risk of recurrence of the disease, lengthy and repeated hospitalizations, etc., must be considered in contemplating the effects on the adjustment of the patient and family.

Despite what would appear to be a predictably traumatic psychological insult to a developing child, the existing empirical evidence would suggest that, overall, the majority of children and adolescents cope relatively well. Indeed, pediatric amputees have been found to achieve generally good functional outcomes.⁶ Nonetheless, some children who undergo limb loss have difficulty coping with various aspects of their experience. However, relatively little has been written concerning ways to facilitate coping and adjustment of these patients and their families to the unique stressors associated with the loss of a limb. The present paper will review the psychological impact of limb amputation on the child and highlight specific approaches to intervention.

Determinants of Adjustment

Multiple factors play a role in determining the child's reaction to amputation. In a recent major review of the literature on the psychosocial effects of limb deficiencies in children, Tyc⁶ implicated the following variables as significant risk and protective factors: the child's developmental level; the child's temperament; level of general stressors and microstressors; coping style and ability; cognitive capacity; attributional processes; CNS involvement; and premorbid functioning. Additional factors may include past experiences with medical treatment and hospitalization, parental reaction and coping style, the nature and extent of functional impairment and physical disfigurement associated with the injury or illness, and the individual personality of the child. Recent empirical research has also suggested that a number of mediator variables in the child's environment may regulate the child's coping and adaptation. These include perceived social support from parents, teachers, and classmates and level of family/marital conflict and organization.^{6,8} These factors impact on the child's self-esteem and perception of his/her physical appearance, which in turn may influence coping and adjustment to the loss entailed with amputation.

Limb amputation has been conceptualized from a psychosexual developmental perspective as potentially impacting upon the normal fears and guilt associated with oedipal strivings and resultant castration anxiety. It has been observed that, in particular, the surgical

intervention of amputation can damage the child's sense of bodily integrity and activate fears of being attacked, overwhelmed, or castrated.⁹ In the older child or adolescent, the loss of a significant body part may reactivate fears of being punished for the fantasized longings of an earlier stage of development.

It is well-documented that the hospitalization experience per se may have an adverse effect on the child.¹⁰ Cognitive and social development factors will influence the way in which the child may perceive and interpret the unfamiliar hospital environment, separation from parents and the security of home routines, and the frequent uncontrollable intrusions of strange people, sights, smells, and sounds, as well as potentially painful and invasive procedures.

Developmental factors play a role in determining adjustment to hospitalization and surgical intervention. The importance of attachment to a primary caregiver and the development of a sense of basic trust makes the hospital environment one in which the infant may be at risk for disruptions in cognitive and affective development, and the formation of healthy attachments. Preschool children, whose newly acquired physical and cognitive skills lead to increased independence and autonomy, may react to the trauma of hospitalization and surgery by regressing to more dependent and immature behaviors, such as bedwetting and thumbsucking. Their egocentric perspective may lead them to perceive aversive experiences as punishment for real or imagined misdeeds. Latency-aged children may react with aggressive behavior and poor compliance to aspects of care which may complicate recovery and rehabilitation. The preoccupation of adolescents with physical appearance and their intense desire to not appear different from their peers may make them particularly prone to experiencing difficulties in coping. Adolescents' efforts at developing a clear sense of self and striving for independence are often at odds with the loss of a sense of personal control associated with illness, failure of bodily functions, and hospitalization. They may react by adopting a "tough" or indifferent facade. Surgery and amputation may be perceived as a particularly brutal assault upon their physical self and, subsequently, upon their bodily integrity, self-esteem, and identity.

Parental and family reaction to the child's illness, hospitalization, and the amputation may serve to mediate the child's reaction and adjustment.¹¹ The provision of support and clear communication of information about all aspects of care may reap rewards in terms of indirectly decreasing the child's anxiety and the potential for maladaptive behavior. Thus, it is crucial to assess the history of parents and family in coping with various life stressors.

Individual child characteristics and the child's history of coping with previous stressful experiences, to include illness, hospitalization, and invasive procedures, will also influence coping and adjustment. Temperamentally, the more "inhibited" or "difficult" child is likely to respond to novel situations with a good deal of anxiety and fearfulness, while the more "uninhibited" or "easy" child may react with more flexibility and adaptability to the threat of hospitalization and impending medical procedures.^{12,13} As Varni et al.⁷ have noted, the child's temperament becomes a factor in the context of the family environment, with more cohesive families apparently able to adjust and compensate for the effects of the child's emotionality on coping.

Lastly, the specific nature of the child's illness or injury, i.e., trauma, disease, or congenital causes, that necessitates amputation, as well as the nature and extent of the surgery and limb loss may determine to some extent the child's reaction. Clinical experience suggests that the greater the number of surgeries or procedures the child must endure, the greater the risk of psychological distress. A child undergoing a single surgery to remove a limb is probably at lower risk emotionally than a child who is subjected to repeated surgeries performed in the hope of saving a limb that must eventually be removed. The child's and family's hopes may be raised and then dashed multiple times, with the accumulation of stress with each subsequent surgery. In addition, the more extensive and disfiguring the amputation, the greater the degree of accommodation to body image changes, as well as the greater the rehabilitative effort.

Coping and Intervention

The well-documented stages of reaction and coping with grief and loss are applicable to the child experiencing amputation.¹⁴ Briefly, the child and family can be expected to experience several phases of psychological stress, including shock, denial and panic, protest and regression, oppositionality, mourning, and readjustment. Similar to adults coping with grief, children do not necessarily move through these stages in any particular sequence. These different phases may be accompanied by specific behaviors and feelings, such as extreme fear or avoidance, which may impair compliance with the child's required medical regimen. However, some denial may be desirable and the skillful clinician may ally with the patient and family in facilitating the therapeutic use of such defenses. This might entail, for example, encouraging a degree of denial by helping the patient focus or

distract himself or herself from a painful procedure. Furthermore, it is important to keep in mind that the patient's family members are concurrently experiencing movement through similar stages in their attempts to cope with the child's amputation, though at their own sequence and timing. For instance, the child may be grieving the loss of the limb while the parent is still in a state of denial. Intervention should be tailored to account for these discrepancies.

Concepts of Coping

The individual's efforts to cope with a stressor have been characterized as falling into two domains: "problem-focused coping" and "emotion-focused coping."¹⁵ Problem-focused coping entails mobilizing the individual's personal and social resources to act upon a particular stressor. "Emotion-focused coping" describes the process by which the individual attempts to alter the various emotional states which are associated with the stressor. Both the individual's efforts at coping with the stressor, as well as the therapeutic interventions employed can conceptually be understood as affecting one or both of these coping domains. For example, the child anticipating an amputation secondary to an organic condition, e.g., malignant osteogenic sarcoma, may attempt to cope with her fear and anxiety by actively withdrawing from health care professionals (problem-focused) and thus denying and avoiding the stressor (emotion-focused). Intervention in such a case might involve coaching and role-playing with the child ways of asking the physicians and nurses for specific preparatory information prior to a procedure (problem-focused) combined with reducing the child's fear through reframing the stressor as helpful in the long run. This could be accompanied by teaching the child strategies for relaxation and imagery/distraction during a painful procedure (emotion-focused).¹⁶

Depending on the nature of the child's amputation, e.g., due to traumatic injury or secondary to malignancy, the anticipatory time available, developmental status, and the child's and family's psychological and biological (i.e., temperament) preparedness to respond to stress, a variety of interventions may be available to the clinician and applicable to a particular child's situation.

Intervention Strategies

While not the primary focus of this paper, it is assumed that a thorough assessment will be conducted prior to intervening with the child

and family experiencing difficulty in coping with amputation.¹⁷ This should include obtaining a thorough health and medical history, assessing the child's developmental and cognitive status, personality, and social skills assessment,¹⁸ and family systems and social support assessment.¹⁹ Various methodologies have been devised for assessing children's adjustment to hospitalization,²⁰ pain,^{21,22} and general health concepts,²⁴ as these factors may be of particular relevance in certain circumstances. Particular attention should be paid to linking the child's developmental level to specific treatment strategies as this has been found to affect both the child's symptomatic presentation,²⁵ as well as the appropriate expectations for the child.²⁶

Play Intervention

Play techniques have a long history as the primary form of individual child intervention in helping children cope with a host of adjustment problems.²⁷ Children's play has been described as the "work of worrying" and is considered to be the child's modal method of self-expression.^{28,29} As Rae³⁰ has summarized, the use of therapeutic play within the hospital setting has been credited with helping children "regain control, express feelings and emotion, gain information about hospital procedures, prepare for medical events and transform children from passive sufferers into active agents in their own care" (p. 618).

Play interventions often appear deceptively simple and unsophisticated. However, most play approaches include elements of a number of intervention techniques, such as role reversal, attention-distraction, imagery, modeling, behavioral rehearsal, and positive reinforcement. Several of these intervention techniques will be discussed below with special emphasis on their use with the child dealing with amputation. Play approaches also place considerable emphasis upon the rapport within the clinician-child relationship as being the primary prerequisite to successful intervention.

Play intervention utilizing dolls and puppets has a long history in helping children cope with many aspects of illness, traumatic injury, and hospitalization. Such play is often used in an informational vein to help prepare the child for what to expect during the hospital stay, as well as for specific treatment approaches, such as surgery,³¹ bone marrow aspirations,³² and other invasive procedures. The use of dolls and puppets provides not only a familiar and nonthreatening vehicle, but also allows the child to become less defensive and more receptive

to information. The child can often readily identify with the vulnerable role of the doll or puppet as patient. This is facilitated when the doll "verbalizes" many of the beliefs and feelings that the child is experiencing. In addition, less directive play with dolls and puppets can provide the child with an outlet for expressing his or her feelings of rage, hostility, and fearfulness associated with being in the dependent and vulnerable role of patient. This approach can provide the clinician with opportunities to correct any misconceptions the child may harbor regarding specific issues pertaining to his or her illness or treatment. In addition, the high degree of age-appropriate egocentrism associated with childhood typically results in significant feelings of guilt and fear of retaliation for real or imagined wrongdoings. A major task of psychotherapeutic intervention with the pediatric amputee is to help alleviate these feelings and to facilitate the child's feeling more in control of his or her situation.

Healy³³ reported on his treatment of a severely regressed four-year-old boy who was anticipating a leg amputation due to gangrene secondary to a traumatic injury. He painted a male doll's leg to resemble the child's preoperative injuries, to which the child initially responded with seeming indifference. However, the child eventually began to show more interest in playing with the doll, which was then used to demonstrate various medical procedures prior to their administration to the patient. The child began to feed and nurture the doll in play and was much more accepting of injections when they were administered first to the doll. Following the child's limb amputation, the doll's leg was also "amputated" in a similar fashion to the child's leg, with the stump bound in bandages. An identical doll was then fitted with a prosthesis and presented to the child several weeks prior to his receiving his own prosthesis. Eventually, the child spontaneously relinquished the original doll and began playing with the doll with the prosthesis, which was felt to be instrumental in his smooth adjustment to his own new prosthetic leg.

Filmed Modeling

Having the pediatric patient view a film of another child who is undergoing a similar procedure has been found to reduce the child's anxiety concerning surgery, both pre- and post-operatively.³⁴ A number of videotaped presentations for use with clinical populations have been developed.³⁵ Such interventions typically depict a child encountering various experiences that are associated with surgery, such as

admission procedures, talking with the surgeon and anesthesiologist, receiving routine lab tests, being shown various types of hospital equipment, being separated from parents, and viewing the operating and recovery rooms. Filmed interventions have been found to be most effective if the child model in the film shares characteristics in common with the patient, i.e., of similar age and of the same sex.³⁶ In addition, this intervention approach is further facilitated if the child in the film narrates the observed procedures, along with providing his or her own thoughts and feelings at critical points. Positive coping self-statements such as "I know I can do it", as well as having the child model positive coping behaviors (relaxation, slow-chest breathing, use of imagery, etc.) and other variations have been added.²¹ Unfortunately, one limitation of this approach is the prohibitive cost of producing the wide variety of films required to meet the needs of the varied populations of patients and conditions. However, the generally lower cost of modern video equipment may facilitate more ready access to this treatment tool.

Emotive Imagery/Relaxation/Hypnosis

A number of intervention techniques have been found to be effective with a variety of painful procedures and medical conditions. These involve application of such techniques as progressive relaxation³⁷, breathing exercises³⁸, emotive imagery^{39,40}, hypnosis^{41,42}, and distraction techniques. These have been found to be effective with a variety of painful and anxiety-producing conditions and procedures. By extension, these techniques appear to have ready applicability with children who are anticipating or have undergone limb amputation, although empirical data regarding their use with this population has yet to be established. Such interventions often serve as an active attention-distraction technique during specific painful procedures, as well as help the child to reframe his or her experience in a more positive light. In addition, creation of a relaxing condition facilitates the child's entrance into a physiological state that is antagonistic to pain and distress.

The technique of emotive imagery involves the use of the child's fantasies of hero and super-hero figures in such a way as to weave these images into a story associated with the child's medical situation so as to encourage reframing and mastery.⁴³ Indeed, storytelling has a rich history as a vehicle for transmitting beliefs and attitudes within many cultures and has been shown to be effective in reducing children's anxieties.⁴⁴ Often these stories integrate the child's pain into

the scene in such a way as to transform the meaning of the pain. For example, a male child might imagine himself to be "Batman", with his leg having been injured in a heroic confrontation with an adversary, such as the "Joker". In this way, the child's pain could be interpreted as the price that must be paid for his brave deeds. The child's prosthesis would become a powerful part of his "physical armor". Of course, assessment of the individual child's fantasy life and connecting with his or her "powerful" images enhances the likelihood that the child will accept and utilize the fantasy and imagery.⁴⁵

In a similar fashion, hypnosis has been found to be effective with children undergoing painful bone marrow aspirations.^{41,46-47} Typically, the child is instructed in the use of an initial induction technique, e.g., eye fixation or the TV technique, given instructions for progressive muscle relaxation and slow chest-breathing, followed by suggestions for positive visualization. Posthypnotic suggestions are then given in order to enhance the child's sense of relaxation, well-being, mastery, and coping. Clinical reports in a few methodologically limited studies have been encouraging regarding the value of hypnosis, but remain in need of more controlled investigations.⁴⁸ Obviously, techniques such as relaxation, imagery, and hypnosis share common elements. Further delineation of these variables for clarification as to what may be at the core of their effectiveness remains for future investigation.

Mastery and Behavioral Rehearsal Techniques

These more direct and active techniques are often employed with younger children as part of their play sessions. In such situations, the child may be given a doll or puppet and encouraged to administer a specific procedure to the figure. Children are often given the actual medical equipment or a "safe" representation and instructed in the procedure. During the child's administration of the procedure, he or she is encouraged to coach the doll or puppet to actively cope, e.g., breathe deeply, distract itself, engage in positive imagery, etc. Role reversal may then ensue wherein the clinician pretends to administer the procedure and the child rehearses the active coping skills. Obviously, for the child who is undergoing amputation, details of the actual surgery may be too overwhelming for explicit depiction in play and behavioral rehearsal. However, more general coping skills may facilitate mastery and help desensitize him or her through exposure under supportive and relaxing conditions.

A multicomponent cognitive-behavioral intervention "package" was

developed and studied by Jay and her colleagues.²¹ This program was designed to teach effective coping skills and reduce children's distress during painful bone marrow aspirations. The key components of the "package" included information concerning the nature of the stressful situation, the teaching of active coping skills (deep breathing, imagery/distraction, filmed modeling), and the use of behavioral rehearsal with contingent positive reinforcement for active coping while exposed to treatment-related stressors. A minimal-attention control group and an oral sedation group who received diazepam were included. The results strongly supported the efficacy of the multicomponent package over both medication and attention control conditions. While the medication group demonstrated lower blood pressure readings, there was no other significant effect on child, parent, and clinician-completed measures. The study design did not allow for clarification as to which components of the treatment package contributed most to the effects obtained. Admittedly, this treatment program was designed for children experiencing acute procedure-related pain that was nondisfiguring. However, these strategies would appear to have relevance to helping enhance the child's general coping skills with the more acute short-term stressors associated with the procedures leading up to the actual surgery for amputation. Unfortunately, interventions employing more play-based treatment approaches, as mentioned above, have not been subjected to well-controlled investigation.^{20,49}

Case Illustration

D.W. is a 10-year-old female who was seen in consultation while hospitalized for chemotherapy treatment of her osteogenic sarcoma of the left proximal femur. During her hospital evaluation, it was determined that she would require an above-the-knee amputation of the limb. When the impending amputation was discussed with her and the family, D.W. became extremely withdrawn and regressive in her behavior. After this, she frequently pulled the covers up over her head when health care professionals entered the room. This seemed to be her way of "shutting out" any further information which was shared with her regarding her condition, treatment, and the eventual amputation. Of note, her family history included the recent death of her maternal grandmother due to breast cancer. Prior to the grandmother's death, D.W. had witnessed her severe chemotherapy-induced nausea and vomiting. This may have influenced D.W.'s expect-

tation of these untoward side effects. Consultation was sought with regard to the patient's withdrawal, fearfulness, and depressed affect during the period of time surrounding the diagnostic work-up and initiation of chemotherapy. The goal of intervention was to help facilitate her coping with her current chemotherapy, impending limb amputation, and the fitting of a prosthesis.

During initial visits, any attempts to directly engage the patient in conversation with regard to her current and future medical situation were met with either silence or very brief responses uttered in an almost imperceptible mumble. Efforts at teaching her relaxation and distraction skills were also similarly ineffective due to her withdrawal and refusal to communicate.

When more direct efforts at intervention were met with repeated failure, therapeutic play was introduced via the medium of puppetry. D.W. demonstrated a dramatic response to this modality with an immediate brightening of her affect, increased verbalization, and more outgoing and participatory behavior. She became noticeably less depressed over a period of several days as play intervention allowed her to begin to explore her fears and anxieties. At times, there was a degree of aggression in her play, with various puppets "devouring" body parts of other characters, as well as themes involving deceit and trickery. This was felt to represent the patient's feelings of anger and mistrust, reflecting her belief that the amputation was somehow a form of deception or punishment.

D.W. responded quite readily to instruction in progressive relaxation and slow-chest breathing techniques when modeled by "Miss Piggy", a female puppet figure. Role reversal was employed in having her teach this puppet how to relax during an invasive procedure. For instance, she instructed the puppet to breathe deeply and slowly and to imagine lying on her bed at home while counting the flowers on the wallpaper. The patient was then observed to employ the same technique during her chemotherapy at those times when she became anxious or felt nauseated. She continued to employ this strategy successfully during later hospitalizations for chemotherapy.

With regard to her anticipation of the limb amputation, again puppetry proved to be an effective modality. Themes in her play with animal puppets included biting and chopping off ears and other body parts. Subsequently, she had the animal characters visit a doctor character who fitted them with "replacements" for the missing body parts. In the course of her play, D.W. remarked that these new parts seemed to work as well as the old.

This was a repetitive theme over the course of several weeks of play

therapy. As the time for the actual amputation approached, specific information regarding the various aspects of the procedure was shared with D.W. This was given in graduated increments, designed to minimize anxiety and to work through her fears. Just prior to the surgery, D.W. was noted to cope rather effectively with little evidence of the severely regressive behavior seen previously.

Following the amputation, D.W. initially remained somewhat dependent on her wheelchair. However, after several weeks, she accepted the prosthesis and began ambulating with considerable confidence in a relatively short period of time. Those working with D.W. felt that her manner of coping with the amputation and surrounding events became much more functional following the introduction of the multicomponent program.

Summary

The loss of a limb through amputation represents a potentially major psychological trauma for the child which has implications for his/her developing body image and sense of mastery. For the very young child, nonverbal and metaphorical "safe" alternatives via play modalities and the active facilitation of coping skills are useful clinical tools to impart information, decrease anxiety, and enhance mastery. Assessment should focus on determining family and developmental influences so as to tailor interventions to suit the individual child's specific needs, to include past medical experiences, coping skills, and nature of injury or illness. A multi-modal approach is advocated in therapeutic work with the pediatric amputee both pre- and post-operatively employing a "package" of play and cognitive-behavioral strategies in the context of a supportive relationship for both the child and family, as well as close involvement with the medical and surgical team. These promising interventions await further empirical support as to their specific role with the child experiencing the loss of a limb.

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Special Education Placements of Language-Disordered Children in a Psychiatric Population

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ABSTRACT: Children evaluated by an interdisciplinary team that provided data on psychiatric disturbance, language handicaps, and psychoeducational assessment were placed in special education classes following the complete evaluation. Discriminant analysis as a special case of multiple regression analysis determined that mean reading achievement score, pre-evaluation classroom placement, and to a lesser degree, Performance IQ predicted classroom placement. Overall, correct classroom predictions approached 90 percent. These findings highlight the value of interdisciplinary evaluation to educators and mental health professionals.

Children with all types of handicaps are known to be at risk for the development of psychiatric disorders.^{1,5} In particular this has been shown to be the case for children with speech and language disorders. A review of the literature on the co-prevalence of psychiatric disorders and communication handicaps shows the incidence to vary between 30-80 percent depending on whether the target population is from a speech and language clinic (30-50%) or from a psychiatric facility (50-80%).

In a series of studies, Cantwell and Baker⁶⁻¹⁰ studied the presence of psychiatric disorders in 600 communication disordered children and adolescents referred to a community speech and language clinic. Rutter¹¹ found psychiatric disturbance in 44 percent of a sample of language disordered children, Stevenson and Richman¹² in 59 percent, and Caceres¹³ in 84 percent.

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